



PA-Net: A hybrid architecture for retinal vessel segmentation[☆]

Xuebing Luo^a, Lingxi Peng^{a,*}, Ziyang Ke^a, Jinhui Lin^a, Zhiwen Yu^b

^a School of Mechanical and Electrical Engineering, Guangzhou University, Guangzhou, Guangdong, 510006, China

^b School of Computer Science and Engineering, South China University of Technology, Guangzhou, Guangdong, 510650, China

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ABSTRACT

This paper proposes a hybrid architecture, PA-Net, which amalgamates the strengths of convolutional neural networks and the transformer model to enhance the precision of retinal vessel segmentation. We propose a novel component, the Lightweight Parallel Transformer (LPT), to augment the transformer's adaptability for the task of retinal vessel segmentation. This LPT addresses the shortcomings of standard transformer that are highly dependent on large datasets and computing resources, and can capture long-range dependencies to prevent slender vessels from breaking. Furthermore, we introduce an Adaptive Vascular Feature Fusion module to offset the vascular information loss induced by downsampling layers, thereby enhancing microvessel recognition. The effectiveness of PA-Net was assessed across four distinct datasets: DRIVE, CHASE_DB1, STARE, and HRF, with sensitivities of 0.8284, 0.8570, 0.8813, and 0.8497, respectively. The results suggest that the proposed method outperforms other state-of-the-art alternatives.

1. Introduction

The only vessels in the human body that can be directly examined without intrusive operations are the retinal vessels, which provide important info. Automatic retinal vessel segmentation is crucial for diagnosing eye and systemic diseases, including hypertension, stroke, glaucoma, and diabetic retinopathy [1]. Currently, numerous automatic retinal vessel segmentation algorithms have been proposed by researchers. However, factors such as poor image quality, significant noise, low vessel-background contrast, slender vessels, and numerous irregular microvascular branches have increased segmentation complexity. As a result, retinal vessel segmentation remains a significant challenge.

Deep learning-based retinal vessel segmentation methods have rapidly advanced, with convolutional neural networks (CNNs) significantly improving the precision of retinal vessel segmentation due to their exceptional contextual info extraction and hierarchical feature representation [5,9,10]. However, since the convolutional operation itself is localized, CNNs struggle to capture the long-range dependencies crucial for semantic segmentation. As a result, retinal vessel segmentation methods based on CNNs still face the issue of slender vessel

breakage. To address the limitations of CNNs, researchers have introduced transformers, known for their long-range dependency modeling capabilities [11,12]. Visual transformers have since gained widespread use in computer vision [13–15]. For the retinal vessel segmentation task, visual transformers typically utilize hybrid architectures of convolutional neural network (CNN) and transformer to construct networks by merely introducing standard transformer modules or self-attention mechanism into CNNs [17,18]. Although this hybrid architecture can roughly aggregate the CNN and transformer, it fails to maximize their advantages when applied to the retinal vessel segmentation task. We believe that this strategy has two main drawbacks.

(1) The standard transformer necessitates substantial computational resources [13,15]. Directly integrating the convolution module with the standard transformer significantly increases the model's training and inference time, ultimately reducing its computational efficiency.

(2) Compared to CNNs, transformers excel in extracting long-range dependencies and achieving higher performance limits. However, standard transformers require extensive data support to demonstrate their advantages [12], which is challenging due to the typically small size of fundus image datasets. Furthermore, transformers are less adept at extracting local detail info.

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* Corresponding author.

E-mail addresses: 2802263952@qq.com (X. Luo), manplx@163.com (L. Peng), 269512583@qq.com (Z. Ke), 942784078@qq.com (J. Lin), zhwyu@scut.edu.cn (Z. Yu).

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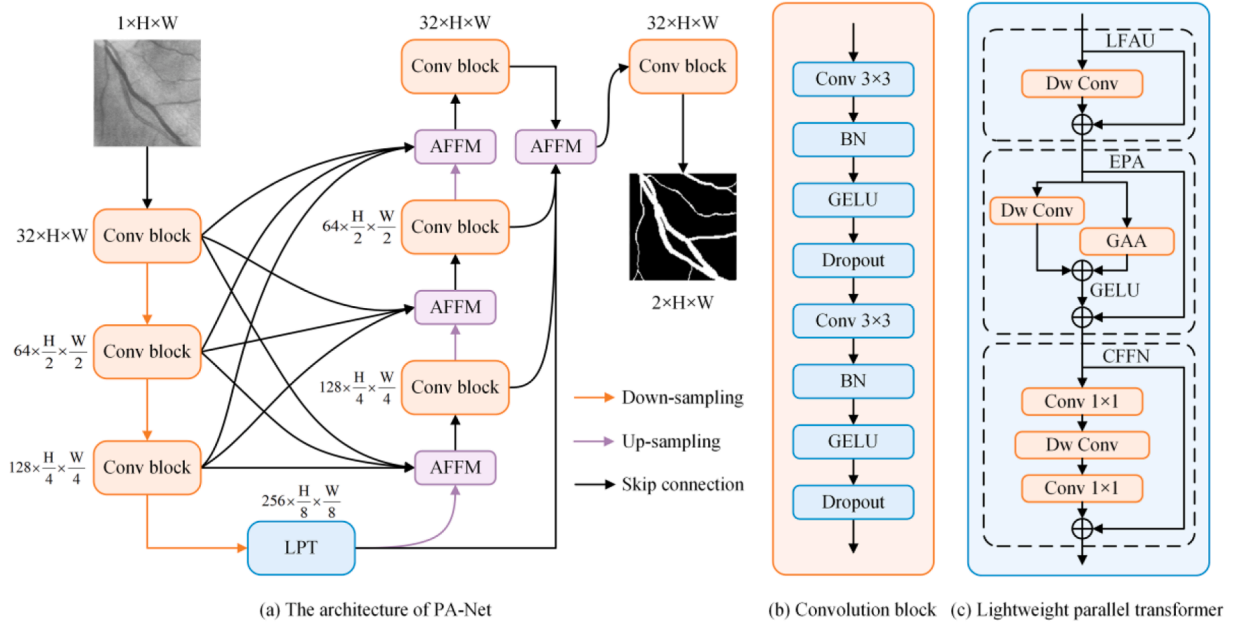


Fig. 1. Overall architecture. (a) shows the architecture of PA-Net, with the three core modules as the convolution block (Conv block), lightweight parallel transformer (LPT), and adaptive vascular feature fusion module (AFFM); (b) is the convolution block's structure; (c) is the structure of the lightweight parallel transformer, where " $\oplus$ " indicates summation, "Dw Conv" means 3×3 depth-wise convolution [25], and "GAA" means gated axial-attention.

To resolve the previously mentioned problems, this paper improves the standard transformer and proposes a lightweight parallel transformer (LPT). LPT is capable of simultaneously extracting long-range dependencies and local detail info required for retinal vessel segmentation while reducing the dependence on computational resources and large datasets. LPT comprises three components: local feature accepting unit (LFAU), efficient parallel attention (EPA), and convolutional feed-forward neural network (CFFN). In this study, we use U-Net [5] as the base architecture and integrate LPT into it to create a novel CNN-transformer hybrid architecture.

The downsampling layers of U-Net result in the reduction of spatial info in the feature map, particularly causing the deep feature map to lose a large amount of spatial info. While the U-Net's skip connection mechanism can mitigate the loss of retinal vessel spatial info to some extent, the final vessel prediction map still suffers from insufficient microvessel recognition [8,27]. We contend that, in contrast to other medical image segmentation tasks, retinal vessel segmentation places a greater emphasis on spatial info retention because it requires the identification of numerous irregular microvessels, a requirement that the skip connection mechanism in U-Net cannot meet. To address this issue, this paper introduces an adaptive vascular feature fusion module (AFFM) to replace skip connections. AFFM can adaptively fuse vascular semantic information at different levels to obtain a more comprehensive semantic expression. The main contributions of this paper are as follows.

(1) Based on the peculiarities of the retinal vascular segmentation task, we propose LPT as an improvement to the standard transformer. LPT can simultaneously extract the long-range dependencies and local detail info required for retinal vascular segmentation, while reducing the dependence on computational resources and large datasets, making it more suitable for the retinal vascular segmentation task.

(2) We propose AFFM to further enhance the utilization of feature mapping at different levels, minimize spatial info loss, and obtain a more comprehensive contextual semantic representation for higher accuracy in vascular prediction maps.

(3) we present a novel hybrid architecture retinal vessel segmentation method, PA-Net, which is based on LPT and AFFM. We implemented comprehensive experiments, and PA-Net achieved state-of-the-art performance on four public datasets, namely DRIVE [19],

CHASE-DB1 [20], STARE [21], and HRF [22].

2. Related work

2.1. Early methods

Retinal vessel segmentation approaches are primarily classified into unsupervised and supervised methods. Unsupervised methods can be further categorized into four groups: morphological and mathematical model methods, matched filter methods, vessel tracking methods, and variable model methods [2]. Supervised approaches can be classified into two categories: traditional machine learning-based methods and deep learning-based methods [4]. Compared with unsupervised methods, supervised methods train network models using manually labeled fundus datasets, learning retinal vessel extraction rules to optimize segmentation models with better segmentation results. Due to their excellent segmentation performance, deep learning-based methods have taken the lead.

2.2. Convolutional neural networks

Deep learning-based image segmentation methods have advanced rapidly, demonstrating remarkable performance in medical imaging tasks like retinal vessel segmentation, with most of the segmentation networks being CNNs. In 2015, Ronneberger et al. proposed U-Net [5], featuring a distinct encoder-decoder structure and skip connection mechanism. Many researchers have combined dense connection mechanisms, attention mechanisms [6], and expansion mechanisms [7] based on U-Net to propose a series of U-Net-derived networks that have demonstrated strong performance in retinal vessel segmentation task. Li et al. developed AFNet [8], which added a positional attention mechanism to U-Net, improving microvascular branch segmentation accuracy. Wang et al. combined residual blocks and dilated convolution to propose AR-SA U-Net [9], while Li et al. [10] introduced a bidirectional attention structure to capture global dependencies and lessen intra-class inconsistencies, effectively identifying irregular retinal vessels. However, due to the local nature of convolutional operations, these CNNs often struggle to capture long-range dependencies. Particularly when dealing

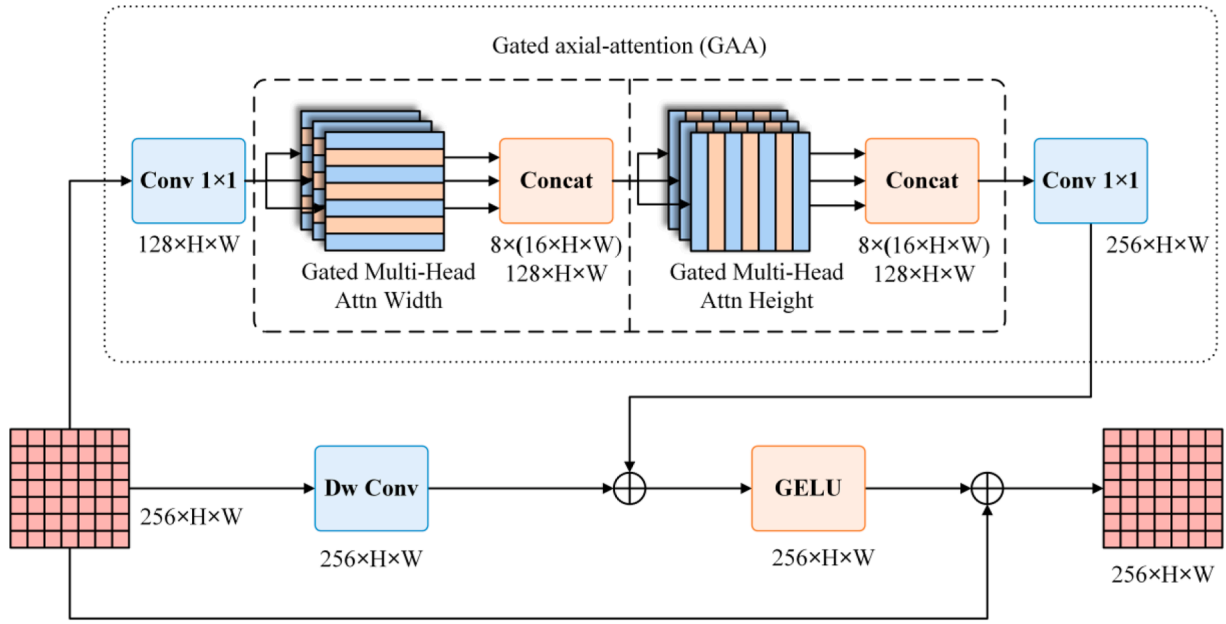


Fig. 2. Efficient parallel attention. Gated axial-attention (GAA) consists of two gated axial-attention layers operating along the width and height axes sequentially. We employ $N = 8$ attention heads. "Dw Conv" means 3×3 depth-wise convolution.

with the task of segmenting slender vessels, CNNs may fail to accurately identify the global vessel structure, leading to the issue of vessel breakage.

2.3. Visual transformer

Transformer [11] was recently introduced to the computational vision from the area of natural language processing owing to its natural and effective modeling capabilities for global characteristics and long-range dependencies [12]. Subsequently, numerous visual transformer-related networks have been proposed, including DAE-Former [14] applied to medical image segmentation, Swin Transformer [13] focusing on model lightweight, and BiFormer [15] and CMT [16] combining CNN and transformer. Scholars generally use the combination of CNN and transformer to build segmentation networks. Wang et al. proposed DA-Net [17], which uses a double-branch transformer module to aggregate global info and local info. To broaden the receptive field and boost the precision of retinal blood vessel segmentation, Han et al. embed the self-attention mechanism into U-Net and suggest NoL-UNet [18]. The above two networks [17,18] simply combine the standard transformer and the convolutional module or introduce the self-attention mechanism into the CNNs. They do not improve the standard transformer following the requirements of the retinal blood vessel segmentation task, which will lead to the defects of the network dependent on computing resources and large datasets, while the fundus datasets are generally small, thus affecting the performance release of the network. Therefore, we propose an optimized hybrid architecture—PA-Net, which can take full advantage of the strengths of CNNs and Transformers, while compensating for the shortcomings of Transformers as much as possible.

3. Proposed method

In this section, we present our proposed method in detail: the overall architecture of PA-Net is introduced in Subsection 3.1, LPT is explained in Subsection 3.2, and AFFM is covered in Subsection 3.3.

3.1. Overall architecture

We suggest a hybrid architecture, PA-Net, that combines the strengths of convolutional neural network and transformer to achieve high-precision retinal vessel segmentation. PA-Net can effectively capture the long-range dependencies and local detail information required for retinal vessel segmentation, while maintaining a low computational cost. In Fig. 1(a), we present the overall architecture of PA-Net, which adopts a classical encoder-decoder structure akin to U-Net. It comprises five main components: convolution block (Conv block), lightweight parallel transformer (LPT), adaptive vascular feature fusion module (AFFM), downsampling, and upsampling. PA-Net first performs feature extraction using the convolution block in the encoder part, and performs three times downsampling using maximum pooling to extract the semantic feature maps at various levels. The deepest feature maps are then input into LPT to capture long-range dependencies and local detail info. We utilize a single LPT embedding strategy, placing a single LPT in the network's deeper layers, reducing computational effort since the computational effort of the LPT is proportional to the resolution of the feature map. Additionally, as the retinal vessel segmentation task requires identifying numerous microvessels and the LPT has limited local detail extraction ability, the single LPT embedding strategy is more suitable for this task. In the decoder part, upsampling and convolution blocks are used to recover the image resolution, and the upsampling is performed by bilinear interpolation. Meanwhile, the semantic info at different levels in the encoder is adaptively fused using AFFM to obtain a comprehensive contextual semantic representation. Then it is passed to the decoder and fused with the decoder's feature map, which in turn compensates for the reduction of spatial info due to multiple downsampling in the encoder and restores the detailed info of the semantic feature map. Then the different levels of vascular semantic info in the decoder are adaptively fused using AFFM. Finally, a convolutional block is passed to output a high-precision retinal vascular prediction map.

3.2. Convolution block

Fig. 1(b) depicts the convolution block's structure. The convolution block is mainly responsible for extracting feature info, and cooperating with downsampling and upsampling to get different levels of vascular

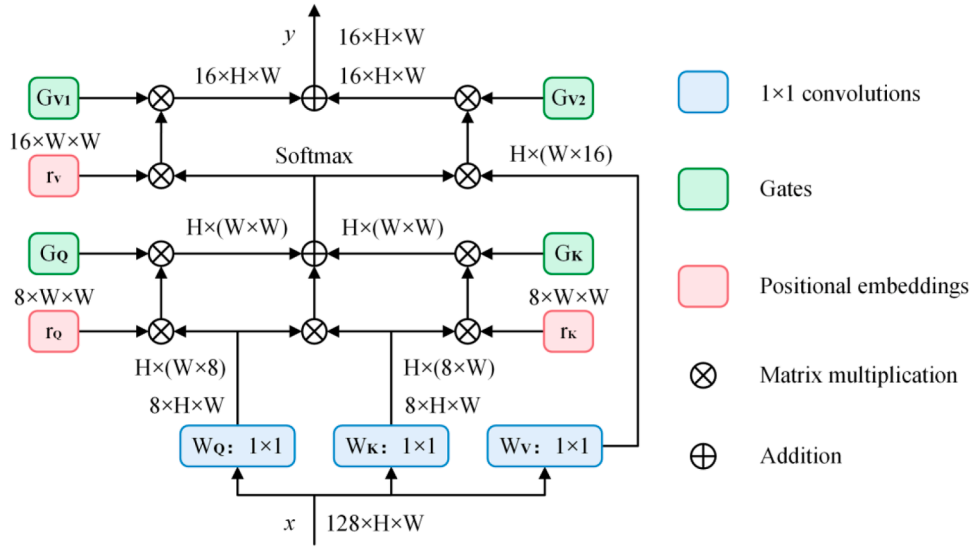


Fig. 3. Self-attention mechanism along the width axis. We employ $N = 8$ attention heads. This diagram shows the calculation process of a single attention head.

semantic feature maps. The convolution block requires less computation and can build the network together with LPT to play a complementary role. The convolution block mainly consists of 3×3 convolution (Conv 3×3), batch normalization (BN), gaussian error linear units (GELU) [28], and Dropout. BN and Dropout can help the model avoid overfitting and improve its capacity for generalization. The GELU function, which is superior to the traditional RELU function, can accelerate the model's convergence.

3.3. Lightweight parallel transformer

We show the structure of the LPT in Fig. 1(c). Retinal vessels are slender and span a large area with many irregularly divided microvascular bifurcations. So the retinal vessel segmentation network needs to focus on not only capturing global contextual info and long-range dependencies, but also extracting local detail info to avoid the problems of slender vessel breakage and vessel discontinuity. Containing a large number of depth-wise convolutions [25] in the LPT, the network can extract both long-range dependencies and local detail info required for retinal vessel segmentation. Containing a great deal of depth-wise convolution in LPT can reduce the dependence of LPT on computational resources and large datasets. Convolutional structures focus on local features and can perform well even on small datasets. Compared with the standard transformer structure, LPT is more suitable for small dataset segmentation tasks such as retinal vessel segmentation. LPT mainly consists of local feature accepting unit (LFAU), efficient parallel attention (EPA), and convolutional feedforward neural network (CFFN), and LPT can be defined as Eq. (1).

$$LPT(x) = CFFN(EPA(LFAU(x))) \quad (1)$$

Where $LFAU(\times)$ is for LFAU, $EPA(\times)$ is for EPA, and $CFFN(\times)$ is for CFFN.

3.4. Local feature accepting unit

Input for the transformer needs to be one-dimensional sequence data rather than two-dimensional picture data, which will destroy the feature map's spatial invariance; although the transformer is good at extracting long-range dependencies, it is not good at extracting local detail info; these are two inherent shortcomings of the transformer when facing vision tasks. To remedy these two shortcomings, we propose LFAU. LFAU consists of 3×3 depth-wise convolution and residual concatenation, which can implicitly embed location encoding info as well as

enhance spatial invariance and extract local detail info [15,16,29]. LFAU can be defined as Eq. (2).

$$LFAU(x) = DwConv(x) + x \quad (2)$$

Where $DwConv(\times)$ denotes the 3×3 depth-wise convolution.

3.5. Efficient parallel attention

We show the structure of EPA in Fig. 2. EPA is the central part of LPT. Compared with the self-attention mechanism, EPA is lighter and more computationally efficient, and can extract long-range dependencies and local detail info required for retinal vessel segmentation. EPA consists of parallel gated axial-attention (GAA) [23,24] and 3×3 depth-wise convolution, GELU activation function, and residual concatenation.

The GAA is responsible for extracting long-range dependencies, the depth-wise convolution is responsible for extracting local detail info, and finally, the fusion of the two semantic info is done by summation operation, and EPA can be defined as Eq. (3).

$$EPA(x) = GELU(Conv(GAA(Conv(x))) + DwConv(x)) + x \quad (3)$$

Where $GAA(\times)$ denotes GAA, $Conv(\times)$ denotes 1×1 convolution, and $GELU(\times)$ denotes the GELU function.

We briefly describe the structure of gated axial-attention in part of Fig. 2. To solve the computational efficiency problem of the self-attention mechanism, it is split into a self-attention mechanism along the width axis of the feature map and a self-attention mechanism along the height axis of the feature map, called gated axial-attention. GAA can be defined as Eq. (4).

$$GAA(x) = Attn_w(Attn_h(x)) \quad (4)$$

The algorithmic complexity of the self-attention mechanism is $O(h^2w^2)$, while the algorithmic complexity of GAA is $O(hwm)$, where $m \in \{h, w\}$. h is the height of the 2D feature map and w is the width of the 2D feature map. Standard self-attention performs correlation calculations on the entire feature map, resulting in a relatively high computational cost. In contrast, gated axial-attention only operates self-attention calculations on the width and height axes of the feature map, needing to calculate the correlation between the pixels and the pixels on its width and height axes, greatly reducing the computational cost.

First, GAA halves the amount of feature map channels using 1×1 convolution. Then computes the self-attention mechanism along the width axis and the self-attention mechanism along the height axis, and utilizes a multi-headed attention mechanism with the number of groups



Fig. 4. Convolutional feedforward neural network.

of 8. Finally, the amount of feature map channels is recovered by 1×1 convolution. To make GAA focus on spatial location info, relative location encoding is added to the query, key, and value respectively, and a gating mechanism is added to the relative location encoding. The position deviations of query, key, and value are easier to learn with large dataset support, while the position deviations are more difficult to learn

when facing a small dataset segmentation task such as retinal vessel segmentation. The performance of the attention itself may suffer if the relative position encoding that was learned is not exact enough. The gating mechanism can control the effects of relative position encoding and thus avoid this situation. Fig. 3 shows the specific structure of the self-attention mechanism along the width axis.

The gating mechanism is expressed as four learnable parameters that can avoid the negative effects of inaccurate relative position encoding. The self-attention mechanism along the width axis can be defined as Eq. (5).

$$y_{ij} = \sum_{w=1}^w \text{softmax} \left(q_{ij}^T k_{iw} + G_Q q_{ij}^T r_{iw}^q + G_K k_{iw}^T r_{iw}^k \right) (G_{V1} v_{iw} + G_{V2} r_{iw}^v) \quad (5)$$

Where q , k , and v are query, key, and value, respectively. r^q , r^k , and r^v denote the relative position encoding of q , k , and v . $G_Q, G_K, G_{V1}, G_{V2} \in \mathbb{R}$

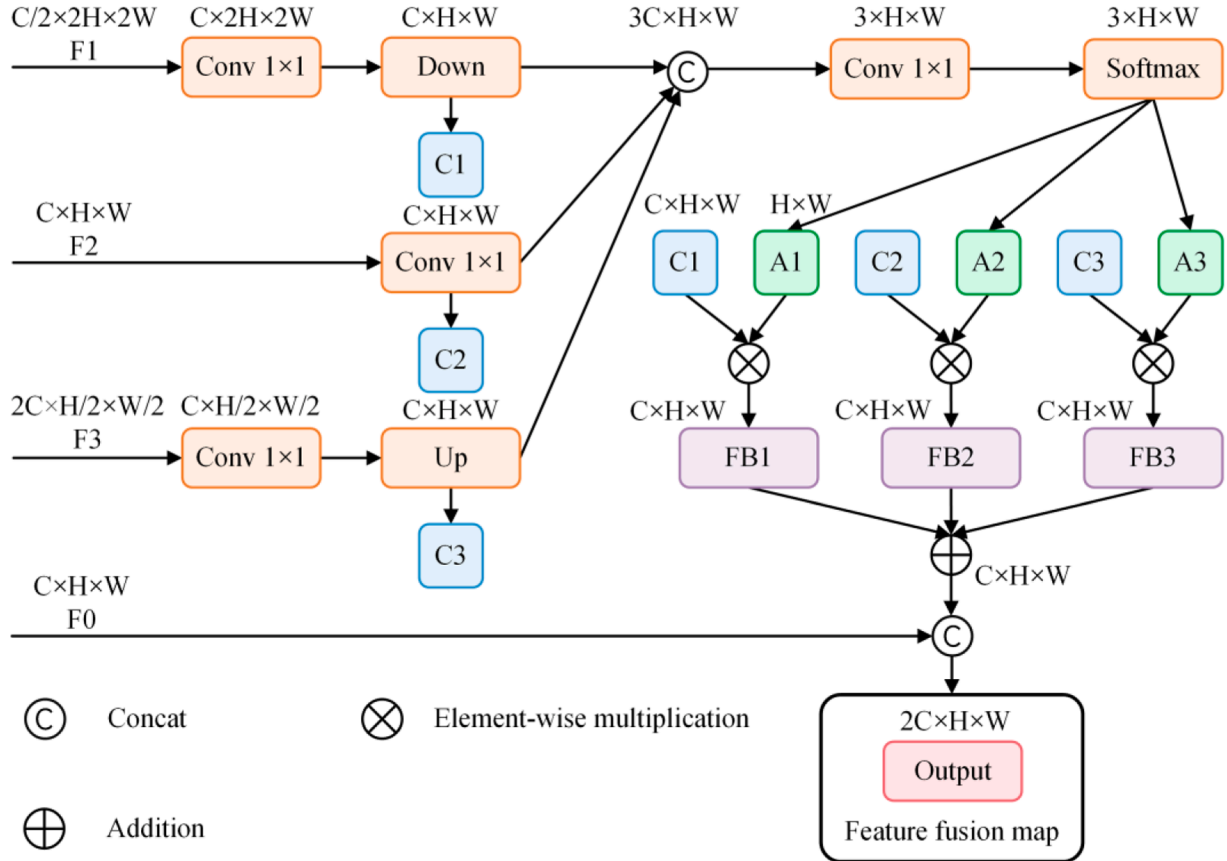


Fig. 5. Adaptive vascular feature fusion module. Down and Up represent downsampling and upsampling, respectively. C1, C2, and C3 represent copies of feature maps. A1, A2, and A3 denote spatial attention weights. FB1, FB2, and FB3 are three fusion branches.

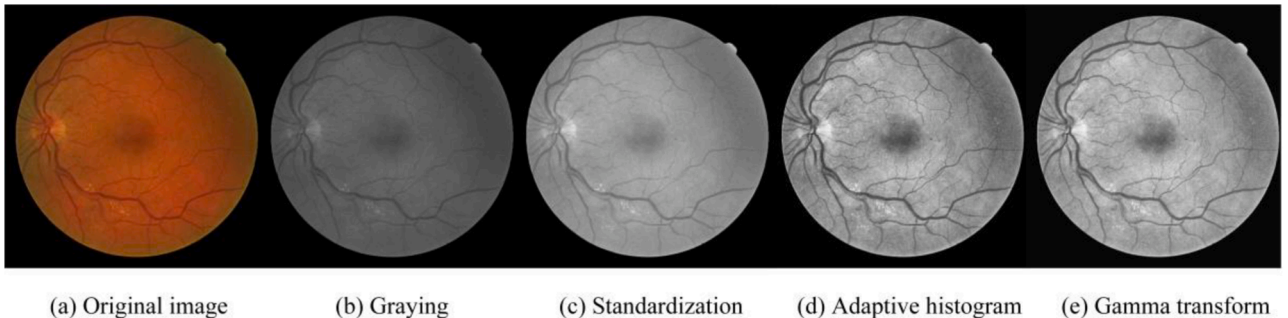


Fig. 6. Image pre-processing effect.

Table 1

Results of ablation experiments on the DRIVE dataset.

Model	SE	SP	ACC	AUC	PR	F1	Parameters	FLOPs	Inference time
U-Net	0.7700	0.9829	0.9548	0.9789	0.8730	0.8183	31.03M	11.51G	231 ms
Baseline	0.7683	0.9801	0.9557	0.9759	0.8591	0.8112	0.91M	2.05G	92 ms
Baseline + MFF	0.7981	0.9776	0.9562	0.9801	0.8593	0.8276	1.33M	3.26G	159 ms
Baseline + AFFM	0.8113	0.9792	0.9577	0.9827	0.8588	0.8344	1.34M	3.32G	169 ms
Baseline + LPT	0.8263	0.9784	0.9572	0.9825	0.8391	0.8327	1.59M	2.21G	162 ms
PA-Net	0.8284	0.9807	0.9582	0.9833	0.8504	0.8393	2.02M	3.47G	189 ms

Table 2

Experimental results on the DRIVE dataset.

Model	Year	Architecture	SE	SP	ACC	AUC	PR	F1	Parameters	FLOPs	Inference time
U-Net [5]	2015	CNN	0.7700	0.9829	0.9548	0.9789	0.8730	0.8183	31.03M	11.51G	231 ms
U-Net++ [31]	2018	CNN	0.7918	0.9828	0.9576	0.9798	0.8758	0.8317	47.19M	28.53G	537 ms
IterNet [30]	2020	CNN	0.7791	0.9831	0.9574	0.9813	0.8694	0.8218	-	-	-
MedT [24]	2021	Transformer	0.8062	0.9789	0.9565	0.9801	0.8514	0.8282	1.51M	1.17G	895 ms
UCTransNet [34]	2022	CNN + Transformer	0.8104	0.9837	0.9593	0.9777	0.8619	0.8354	66.24M	10.80G	635 ms
Wave-Net [32]	2023	CNN	0.8164	0.9764	0.9561	-	0.8346	0.8254	-	-	-
AFNet [8]	2023	CNN	0.8139	0.9818	0.9580	0.9820	0.8394	0.8265	0.66M	1.43G	112 ms
DAE-Former [14]	2023	CNN + Transformer	0.7928	0.9846	0.9592	0.9780	0.8871	0.8373	29.69M	8.53G	370 ms
ResDO-UNet [33]	2023	CNN	0.7985	0.9791	0.9561	-	0.8488	0.8229	-	-	-
PA-Net	2023	CNN + Transformer	0.8284	0.9807	0.9582	0.9833	0.8504	0.8393	2.02M	3.47G	189 ms

Table 3

Experimental results on the CHASE_DB1 dataset.

Model	Year	Architecture	SE	SP	ACC	AUC	PR	F1	Parameters	FLOPs	Inference time
U-Net [5]	2015	CNN	0.8032	0.9789	0.9630	0.9822	0.7916	0.7974	31.03M	11.51G	231 ms
U-Net++ [31]	2018	CNN	0.8214	0.9811	0.9667	0.9869	0.8132	0.8173	47.19M	28.53G	537 ms
IterNet [30]	2020	CNN	0.7970	0.9823	0.9655	0.9851	0.8178	0.8073	-	-	-
MedT [24]	2021	Transformer	0.7901	0.9822	0.9648	0.9849	0.8159	0.8028	1.51M	1.17G	895 ms
UCTransNet [34]	2022	CNN + Transformer	0.8178	0.9798	0.9651	0.9856	0.8013	0.8095	66.24M	10.80G	635 ms
Wave-Net [32]	2023	CNN	0.8284	0.9821	0.9664	-	0.8415	0.8349	-	-	-
AFNet [8]	2023	CNN	0.8194	0.9817	0.9669	0.9867	0.8136	0.8165	0.66M	1.43G	112 ms
DAE-Former [14]	2023	CNN + Transformer	0.8328	0.9792	0.9660	0.9870	0.8001	0.8161	29.69M	8.53G	370 ms
ResDO-UNet [33]	2023	CNN	0.8020	0.9794	0.9672	-	0.8463	0.8236	-	-	-
PA-Net	2023	CNN + Transformer	0.8570	0.9779	0.9677	0.9875	0.8061	0.8308	2.02M	3.47G	189 ms

Table 4

Experimental results on the STARE dataset.

Model	Year	Architecture	SE	SP	ACC	AUC	PR	F1	Parameters	FLOPs	Inference time
U-Net [5]	2015	CNN	0.8251	0.9805	0.9644	0.9832	0.8171	0.8211	31.03M	11.51G	231 ms
U-Net++ [31]	2018	CNN	0.8372	0.9801	0.9658	0.9837	0.8382	0.8377	47.19M	28.53G	537 ms
IterNet [30]	2020	CNN	0.7715	0.9886	0.9701	0.9881	0.8628	0.8146	-	-	-
MedT [24]	2021	Transformer	0.8388	0.9834	0.9689	0.9883	0.8492	0.8440	1.51M	1.17G	895 ms
UCTransNet [34]	2022	CNN + Transformer	0.8426	0.9837	0.9696	0.9885	0.8520	0.8473	66.24M	10.80G	635 ms
Wave-Net [32]	2023	CNN	0.7902	0.9836	0.9641	-	0.8392	0.8140	-	-	-
AFNet [8]	2023	CNN	0.7972	0.9779	0.9712	0.9902	0.8682	0.8312	0.66M	1.43G	112 ms
DAE-Former [14]	2023	CNN + Transformer	0.8266	0.9866	0.9706	0.9897	0.8701	0.8478	29.69M	8.53G	370 ms
ResDO-UNet [33]	2023	CNN	0.7963	0.9792	0.9567	-	0.8392	0.8172	-	-	-
PA-Net	2023	CNN + Transformer	0.8813	0.9805	0.9709	0.9908	0.8323	0.8561	2.02M	3.47G	189 ms

are the control parameters of the gating mechanism. The self-attention mechanism along the height axis possesses a similar structure as the self-attention mechanism along the width axis.

3.6. Convolutional feedforward neural network

We show the structure of the CFFN in Fig. 4. The standard transformer's self-attention mechanism is utilized to obtain spatial correlation, while the multilayer perceptron is used to obtain channel correlation. The multilayer perceptron consists of two fully connected layers. Since the fully connected layer corrupts the spatial info of the feature map, we use two 1×1 convolutions and a 3×3 depth-wise convolution in place of the fully connected layer and propose CFFN.

The expansion ratio of CFFN is 4. CFFN can avoid the destruction of spatial info of feature maps and capture channel correlation, which complements the spatial correlation captured by EPA, and its convolution structure can also enhance the ability of LPT to capture local features [16,26,29].

The CFFN is shown in Eq. (6).

$$CFFN(x) = Conv(DwConv(Conv(x))) + x \quad (6)$$

3.7. Adaptive vascular feature fusion module

The U-Net encoder performs multiple downsampling operations on the feature map, reducing its resolution and revealing hidden semantic info. However, this process also results in a significant loss of spatial

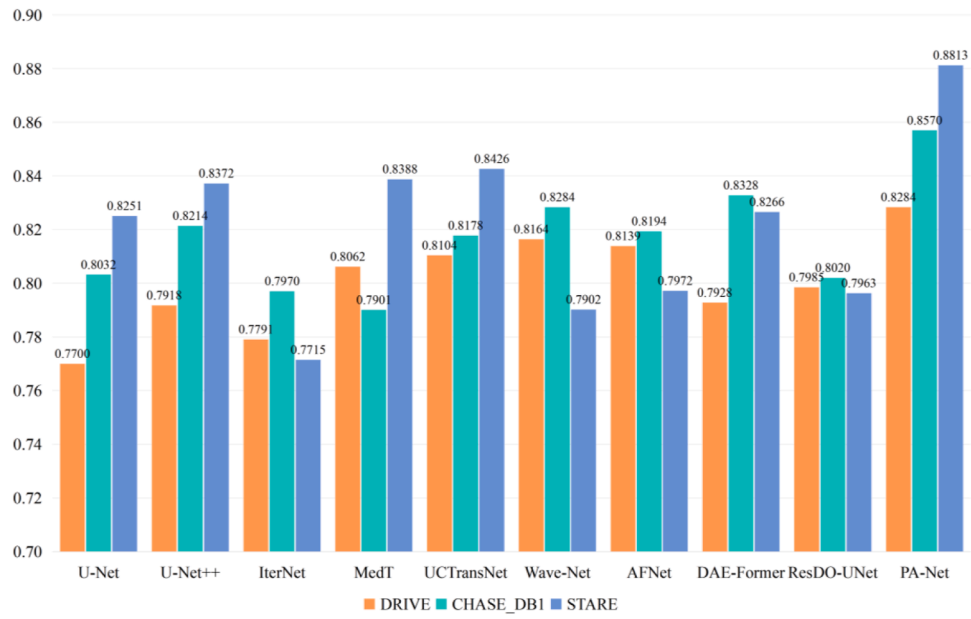


Fig. 7. Sensitivity comparison of different models.

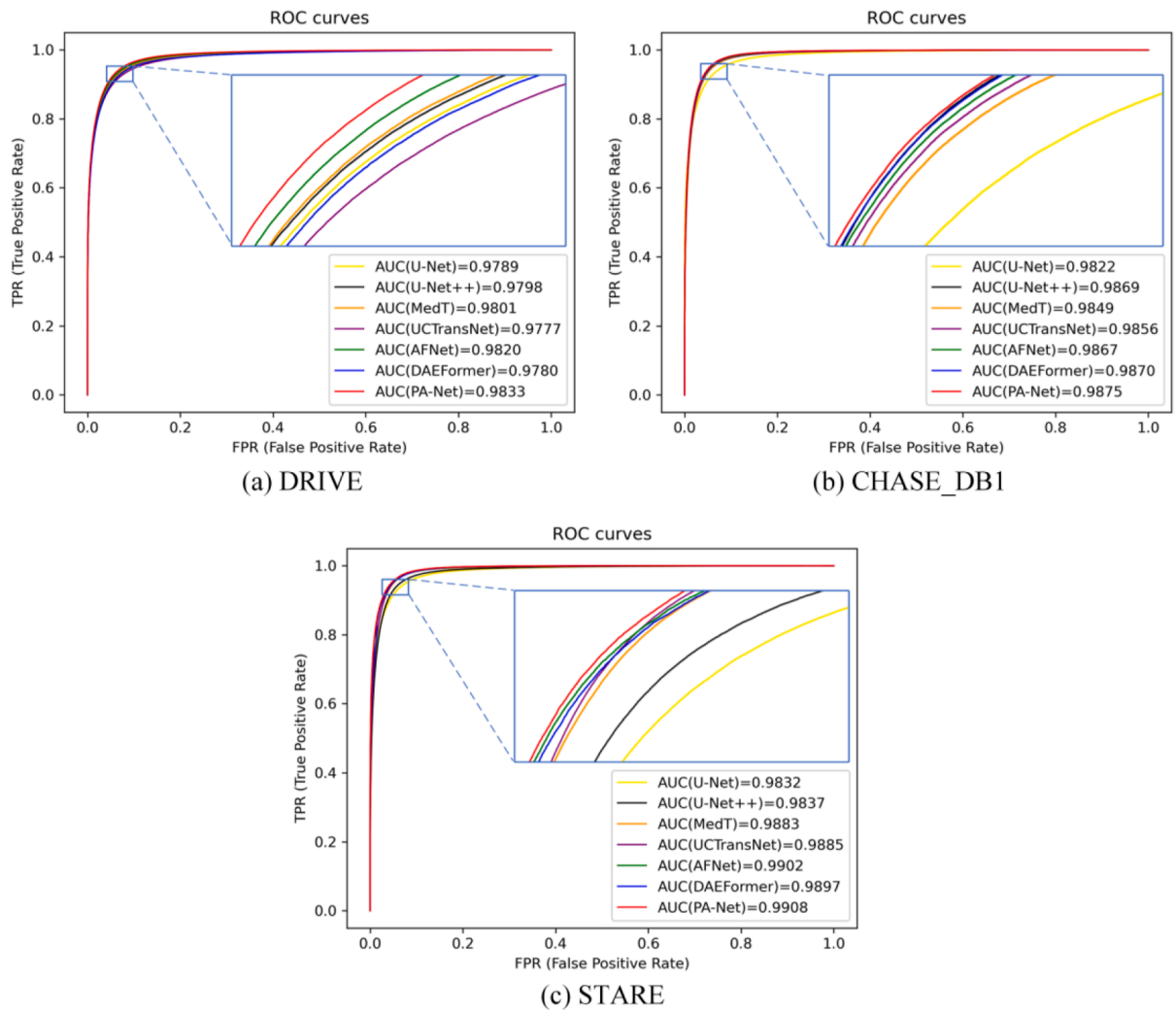


Fig. 8. ROC curve on DRIVE, CHASE_DB1, and STARE datasets.

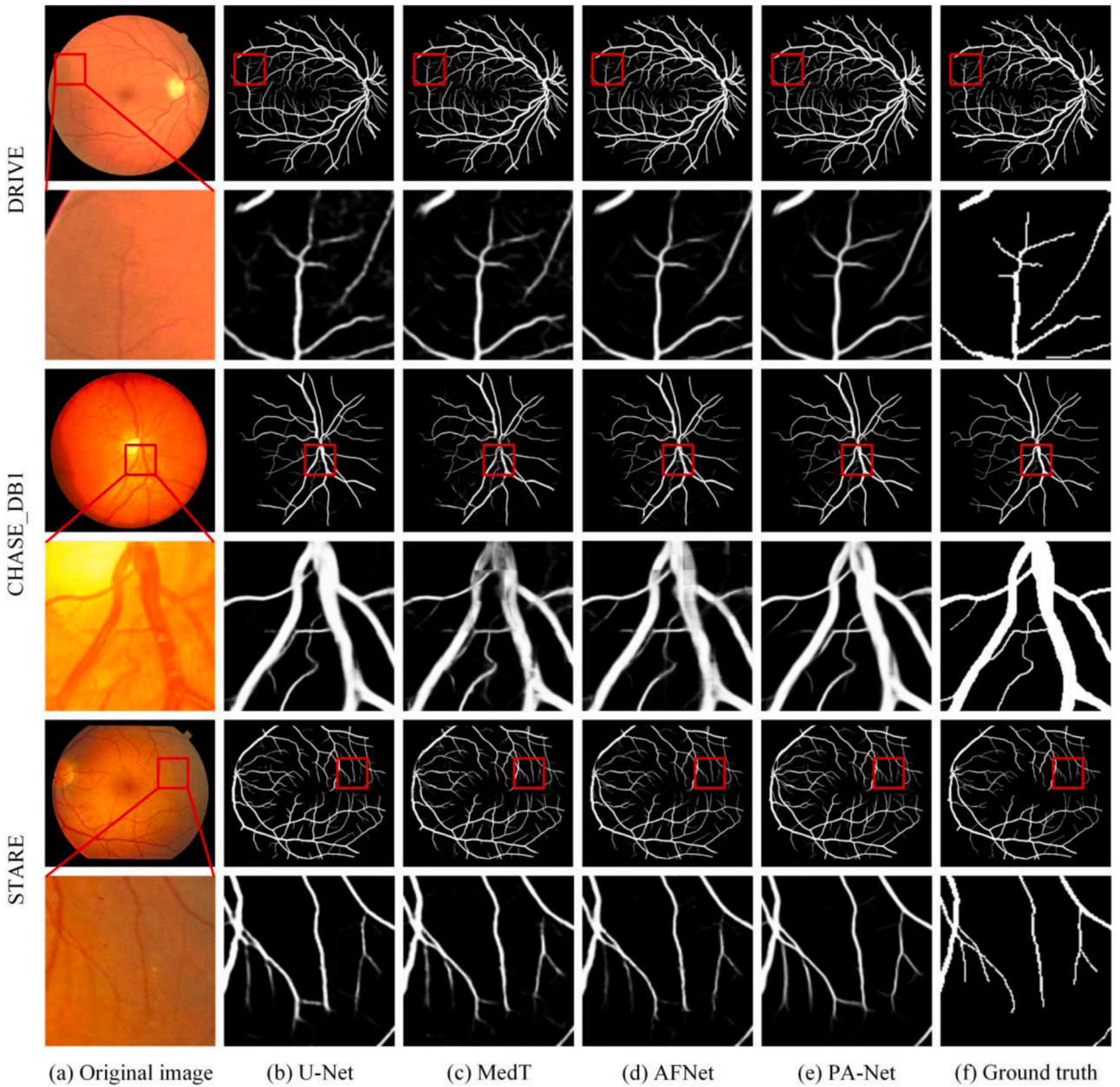


Fig. 9. Segmentation results on DRIVE, CHASE_DB1, and STARE datasets.

info. Relying solely on skip connections to link the encoder and decoder at an identical level cannot maximize the utilization of feature info at various levels or compensate for the lost spatial info. This issue is particularly prominent in retinal vessel segmentation, where a considerable amount of vessel feature info is lost. To fix this issue, we propose AFFM which adaptively fuses deep semantic info with shallow, rich spatial info. The original skip connection mechanism is replaced by AFFM, and the adaptive fusion of vascular semantic info at different levels in the decoder is employed to obtain a more accurate vascular prediction map. Unlike the simple concatenation operation in [8,27], our AFFM utilizes the softmax function to generate multiple spatial weights, guiding the adaptive fusion of feature maps to obtain a more comprehensive semantic representation. This multi-scale module also has the effect of smoothing noise. We incorporate four AFFMs in PA-Net, each with similar structures. Fig. 5 illustrates the structure of one of the AFFMs (the second AFFM from bottom to top), which will be detailed below.

F_0 , F_1 , F_2 , and F_3 are the inputs of AFFM. Firstly, the three feature

maps of F_1 , F_2 , and F_3 are subjected to 1×1 convolution, upsampling, or downsampling operations to make their channel numbers and resolutions consistent, and three new feature maps are obtained: F_1' , F_2' , F_3' . Save copies of these three new feature maps: C_1 , C_2 , and C_3 . The above operations can be defined as Eqs. (7)–(9).

$$F_1' = \text{Down}(\text{Conv}(F_1)) \quad (7)$$

$$F_2' = F_2 \quad (8)$$

$$F_3' = \text{Up}(\text{Conv}(F_3)) \quad (9)$$

Where $\text{Down}(\times)$ denotes downsampling. $\text{Up}(\times)$ means up-sampling. The three feature maps of F_1' , F_2' and F_3' are concatenated, and then the three spatial attention weights A_1 , A_2 , and A_3 are generated by 1×1 convolution and softmax function. This part of the operation can be defined as Eq. (10).

$$A_1, A_2, A_3 = \text{Split}(\text{Softmax}(\text{Conv}([F_1', F_2', F_3']))) \quad (10)$$

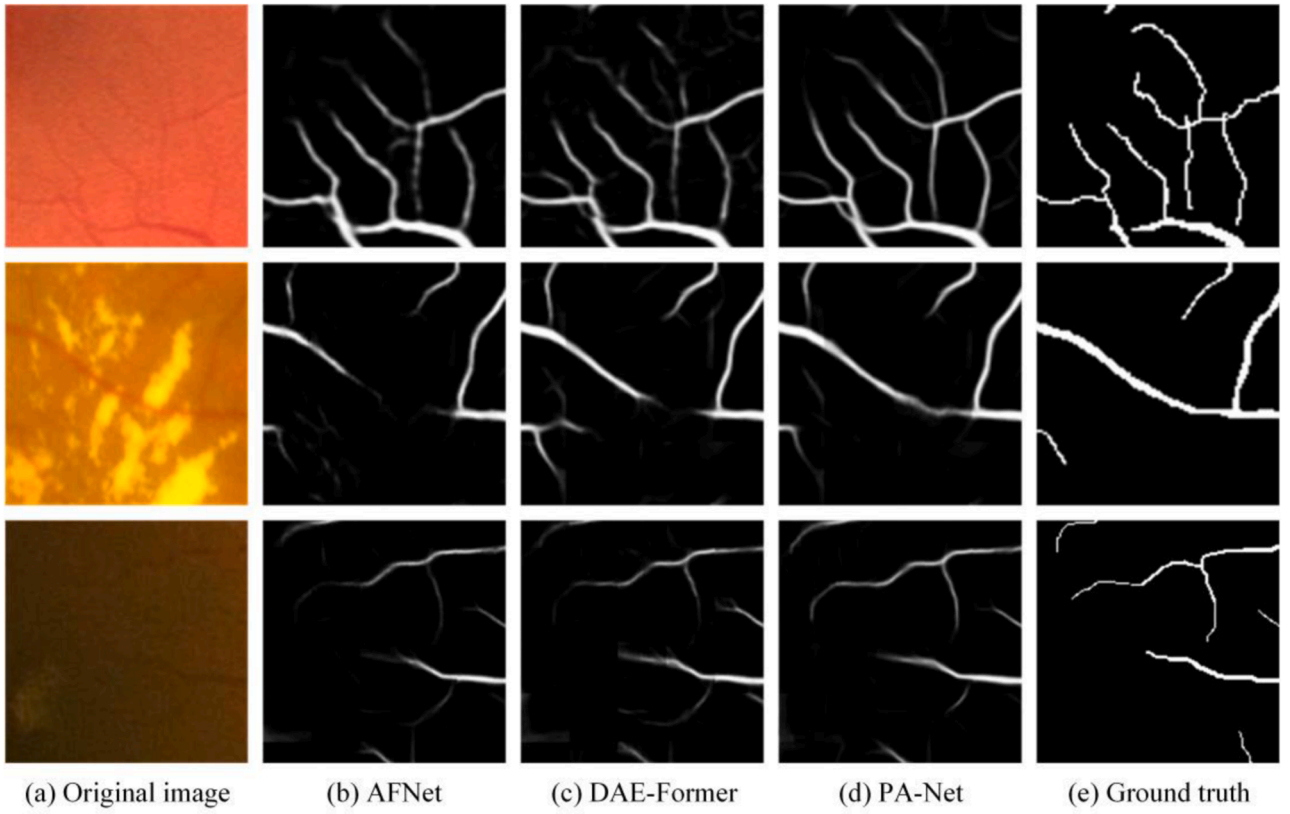


Fig. 10. Anomalies in the datasets.

Table 5

Experimental results on the HRF dataset.

Model	Year	Architecture	SE	SP	ACC	AUC	PR	F1
U-Net [5]	2015	CNN	0.7273	0.9890	0.9657	0.9427	0.8656	0.7904
NUA-Net [3]	2021	CNN	0.8554	0.9741	0.9648	0.9824	-	-
Zhou et al. [35]	2021	CNN	0.8310	0.9730	0.9559	0.9693	0.8115	0.8211
Bridge-Net [36]	2022	CNN	0.8570	0.9690	0.9590	-	-	-
AFNet [8]	2023	CNN	0.8178	0.9810	0.9647	0.9820	0.8035	0.8106
DAE-Former [14]	2023	CNN + Transformer	0.8220	0.9837	0.9667	0.9847	0.8357	0.8288
PA-Net	2023	CNN + Transformer	0.8497	0.9825	0.9682	0.9858	0.8144	0.8317

Where $[,]$ denotes splicing. $Split(\times)$ represents the splitting operation, and the three spatial attention weights $A1$, $A2$, and $A3$ are obtained by splitting. Multiply $A1$, $A2$, and $A3$ with $C1$, $C2$, and $C3$ feature maps respectively, as well as sum up the multiplied results to get the feature fusion map of $F1$, $F2$, and $F3$. Finally, this feature fusion map and $F0$ are concatenated to create the final feature fusion map. This part of the operation is shown in Eq. (11).

$$F_{out} = [(C1 * A1 + C2 * A2 + C3 * A3), F0] \quad (11)$$

Where "*" means Element-wise Multiplication.

4. Experiments

In this section, we conduct experiments on four public retinal fundus image datasets: DRIVE, CHASE_DB1, STARE, and HRF. We evaluate the performance of PA-Net and compare it with other state-of-the-art methods.

4.1. Datasets

DRIVE. The DRIVE [19] dataset has a total of 40 retinal images with

a resolution of 564×584 pixels. Thirty-three of these images were lesion-free retinal images and seven showed signs of diabetic retinopathy.

CHASE_DB1. The CHASE_DB1 [20] dataset has a total of 28 color retinal images. The images were sourced from the right and left eyes of 14 children with a resolution of 999×960 pixels.

STARE. The STARE [21] dataset contains a total of 20 normal fundus images and lesioned fundus images with an image resolution of 700×605 pixels.

HRF. The HRF [22] dataset contains 45 full-color fundus images, each with a resolution of 3504×2336 pixels. To comprehensively evaluate the performance of the model, we use the method of five-fold cross-validation to conduct experiments on these four public datasets.

4.2. Data preprocessing

Enhancing the contrast and lowering noise in fundus images are two goals of data preprocessing. The fundus images after data preprocessing are easier to be extracted by the network model to the blood vessel features, which improves the segmentation precision of the model. The data preprocessing in this paper is divided into four parts: (1) graying, (2) standardization, (3) adaptive histogram equalization, and (4)

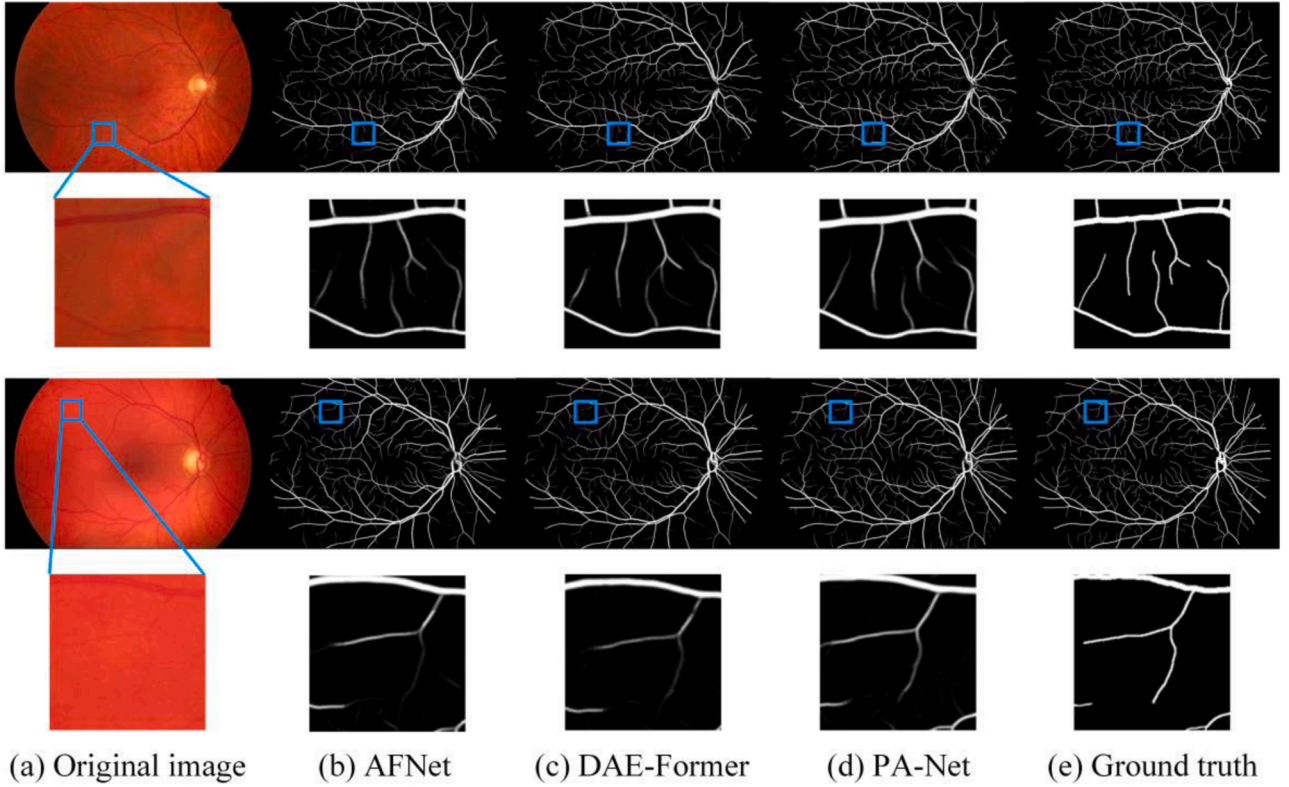


Fig. 11. Segmentation results on the HRF dataset.

gamma transform. Fig. 6 illustrates how data preprocessing has an impact.

4.3. Evaluation metrics

To quantitatively evaluate the models' segmentation performance, we employ six metrics: Sensitivity (SE), Specificity (SP), Accuracy (ACC), Area Under the Receiver Operating Characteristic Curve (AUC), Precision (PR), and F1-score (F1). SE, SP, ACC, PR, and F1 can be defined as Eqs. (12)–(16).

$$SE = \frac{TP}{TP + FN} \quad (12)$$

$$SP = \frac{TN}{TN + FP} \quad (13)$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

$$PR = \frac{TP}{TP + FP} \quad (15)$$

$$F1 = \frac{2 * PR * SE}{PR + SE} \quad (16)$$

The expert manually labeled vessel segmentation images are used as the gold standard. Where TP denotes true positive, TN denotes true negative, FP indicates false positive, and FN indicates false negative. The closer the six evaluation metrics SE, SP, ACC, AUC, PR, and F1 are to 1, the greater the performance of the model is.

4.4. Implementation details

Due to the small number of images and high image resolution of the fundus dataset, the dataset can be augmented. We adopt an image patch input strategy. First, each image after preprocessing is randomly

cropped into multiple image patches of 128×128 pixels. Then image enhancement techniques are used on the image patches to avoid the overfitting phenomenon. The image enhancement techniques used are random horizontal flip, random vertical flip, and random rotation.

The computer configuration used for the experiments is : GPU: NVIDIA GeForce RTX 3090 (24G), memory: 42 GB, and OS: Linux. We use the Pytorch framework to build, train and test the network model. The batch size is set to 16, the initial learning rate is set to 0.0001, the learning rate is iterated using the CosineAnnealingLR strategy, and the loss function is a cross-entropy loss function. The model is trained for 100 epochs using the Adam optimizer. Cross-entropy loss is shown in Eq. (17).

$$L_{ce} = -\frac{1}{N} \sum_{i=1}^N g_i \ln(p_i) + (1 - g_i) \ln(1 - p_i) \quad (17)$$

Where N denotes the total number of pixel points, p_i denotes the probability that pixel i is predicted to be a blood vessel, and g_i is the true label of pixel i .

4.5. Ablation experiments

We performed ablation experiments on the DRIVE dataset to confirm that the LPT and AFFM suggested in this study may effectively increase the precision of the retinal vascular segmentation network. To further validate the effectiveness of AFFM, we use the conventional Multi-level Feature Fusion Module (MFF) [8,27] as a comparison. The main difference between MFF and AFFM is that MFF lacks spatial weights generated by the softmax function. MFF can only simply merge multi-scale feature maps through concatenation operations, and cannot use spatial weights for adaptive feature fusion like AFFM does. Apart from this, MFF and AFFM are similar in other structural aspects and module embedding positions. We use the lightweight U-Net as the baseline, which is a pruned, lightweight model of U-Net. The difference between the lightweight U-Net and U-Net is that the lightweight U-Net

has fewer network layers, and the number of channels in the feature maps of each layer is half that of U-Net. We performed six sets of experiments: (1) U-Net; (2) Baseline; (3) Baseline + MFF; (4) Baseline + AFFM; (5) Baseline + LPT; and (6) PA-Net. The models were evaluated using six metrics: SE, SP, ACC, AUC, PR, and F1. Additionally, we recorded each model's 'Parameters', 'FLOPs', and 'Inference time'. Table 1 displays the trial outcomes.

According to the quantitative analysis based on Table 1, after introducing LPT and AFFM, most evaluation metrics of the baseline model showed significant improvements. This validates the effectiveness of LPT and AFFM. Compared to the experimental group "Baseline + MFF", the experimental group "Baseline + AFFM" had better evaluation scores, demonstrating that AFFM outperforms MFF and further verifying the effectiveness of AFFM. The model's 'Parameters', 'FLOPs', and 'Inference time' only had a small increase after adding LPT, verifying the low dependence of LPT on computational resources. Compared to U-Net, our PA-Net has substantially lower computational consumption while yielding superior segmentation results.

4.6. Comparison with the state-of-the-art methods

We conducted a quantitative comparison of PA-Net with other state-of-the-art methods currently employed for retinal vessel segmentation, performing experiments on three datasets: DRIVE, CHASE_DB1, and STARE. The models were evaluated using six metrics: SE, SP, ACC, AUC, PR, and F1. Additionally, we recorded the models' 'Parameters', 'FLOPs', and 'Inference time'. The comparative results are presented in Tables 2–4.

Upon analyzing Table 2 through Table 4, several useful observations can be made: (1) On the DRIVE dataset, PA-Net achieved the highest SE, AUC, and F1 values. On the CHASE_DB1 dataset, PA-Net attained the highest SE, ACC, and AUC values. On the STARE dataset, PA-Net obtained the highest SE, AUC, and F1 values. In the comparison of 'Parameters', 'FLOPs', and 'Inference time', PA-Net also achieves good results. These results indicate that PA-Net delivers the best comprehensive performance on the three datasets and can accomplish high-precision retinal vessel segmentation. (2) SE represents the proportion of correctly classified vessel pixels to labeled vessel pixels, reflecting the network's ability to identify vessel pixels. In fundus images, the proportion of vessel pixels and background pixels is highly unbalanced, so we consider the SE metric to be relatively more important. The SE metric of PA-Net significantly outperforms other networks, indicating that our method is more proficient in recognizing vascular pixels. Fig. 7 compares the SE metrics of various models in the form of a bar graph, where it can be observed that the SE value of PA-Net significantly outperforms that of other models.

Fig. 8 illustrates the ROC curves (Receiver Operating Characteristic curve) of seven models on the three datasets. The closer the ROC curve is to the upper left corner, the better the model performs. Upon observation, it is evident that the curve of PA-Net is closest to the upper left corner, indicating that PA-Net performs the best under this metric.

4.7. Qualitative analysis

Fig. 9 presents the segmentation outcomes of PA-Net for the three datasets, alongside the results from U-Net, MedT, and AFNet for comparison. The details within the red box are also depicted in the figure with magnification. Several valuable observations can be obtained from Fig. 6: (1) PA-Net exhibits higher segmentation accuracy for slender blood vessels when compared to other networks. There are fewer vessel breaks, and the connectivity of vessels is well maintained. This confirms that LPT effectively captures long-range dependence to prevent vessel breakage. (2) PA-Net is capable of identifying more microvessels and preserving a greater number of retinal vascular spatial structures. It verifies that AFFM can make up for the diminution of spatial info brought on by downsampling layers and retain more info of

microvascular features. Overall, our PA-Net outperforms the other methods in segmentation results across three datasets.

Fig. 10 reveals the segmentation results of PA-Net dealing with anomalies in the dataset, with the segmentation results of AFNet and DAE-Former provided for comparison. The first row displays areas with a large number of microvessels and low contrast, where identifying microvessels may be challenging. The second row presents areas of white specks, where in this scenario, the white specks could interfere with the model's segmentation performance, leading to broken vessels. The third row shows areas with a lot of noise and low contrast, where the difficulty of segmentation is extremely high, easily leading to mis-segmentation and missed segmentation of vessels. Compared with AFNet and DAE-Former, the segmentation results of PA-Net are superior. In the first row, PA-Net successfully identified the contours of the microvessels; in the second row, PA-Net effectively avoided the interference of white specks, maintaining the continuity of the vessels; in the third row, despite facing areas of extremely poor image quality, PA-Net only exhibited a few instances of mis-segmentation. Through the above qualitative analysis, it can be seen that PA-Net performed exceptionally well in dealing with anomalies in the dataset.

4.8. Performance on high-resolution dataset

We evaluated the performance of PA-Net on the high-resolution retinal fundus (HRF) dataset. The HRF dataset contains 45 full-color fundus images, each with a resolution of 3504×2336 pixels. In Table 5, we compared the performance of PA-Net with the current state-of-the-art methods. These comparison data come from the papers we cited and our experimental data. Observing these data, we can find that PA-Net achieved the highest values in ACC, AUC, and F1. In addition, its SE and SP are also very close to the highest values. These results indicate that the overall performance of PA-Net is superior to the comparison methods. Fig. 11 shows the segmentation results of PA-Net on the HRF dataset, with the segmentation results using AFNet and DAE-Former as comparisons. In the details part of the blue box, we can see that PA-Net can well identify the contour of microvessels, and when segmenting slender vessels, PA-Net can also maintain the connectivity of the vessels well. These results indicate that compared to the advanced models used for comparison, our method exhibits better overall performance.

5. Conclusion

Retinal vessels are slender, intricate, and replete with numerous microvascular branches, which presents a significant challenge for the retinal vessel segmentation task. To enhance segmentation accuracy, this paper incorporates the transformer structure within U-Net and proposes a novel hybrid architecture, PA-Net. We improve the transformer structure and propose LPT. This reduces the reliance on large datasets and computational resources while simultaneously extracting long-range dependencies and local detail info. Additionally, we also propose AFFM for the adaptive fusion of various levels of feature maps. AFFM can compensate for the loss of vascular info brought on by downsampling layers and obtain a more comprehensive representation of vascular features. To confirm the efficacy of LPT and AFFM, we performed ablation experiments on PA-Net. Comparing PA-Net with other state-of-the-art methods, experimental results demonstrate that our PA-Net exhibited state-of-the-art performance across all four datasets: DRIVE, CHASE_DB1, STARE, and HRF.

Although we have made good progress, PA-Net suffers from a small number of vessel mis-segmentation problems when processing low-contrast and noisy images. A large amount of image noise can affect the segmentation of the algorithmic model. In the future, we will continue our efforts to find a more reasonable solution to further improve the performance of the retinal blood vessel segmentation model.

CRediT authorship contribution statement

Xuebing Luo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lingxi Peng:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Ziyan Ke:** Validation, Formal analysis, Conceptualization. **Jinhui Lin:** Visualization, Formal analysis, Conceptualization. **Zhiwen Yu:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lingxi Peng reports financial support was provided by Guangzhou Municipality Bureau of Education.

Data availability

The data that has been used is confidential.

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